

Microstructure and physical properties of lime-clayey adobe bricks

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Abstract

Microstructural changes of adobe bricks made of lime-clayey raw material mixes were investigated by X-ray diffraction, infrared spectrometry, differential thermal analyses, scanning electron microscopy and energy dispersive spectrometry. The impact of these changes on the mechanical resistance and water absorption of adobe bricks was evaluated. It is found that lime additions resulted in the development of calcite and poorly crystallized calcium silicate hydrate (CSH). The latter compound is mainly originated from the reaction between fine filiform quartz and lime. The contribution of the pozzolanic reaction, involving kaolinite (the unique clayey constituent of the raw material), to CSH formation seems to be insignificant. The development of CSH together with the formation of minor amounts of calcite, in <10 wt.% lime samples, improved the mechanical strength and the compactness of adobe bricks. However, the quasi-absence of CSH and the excessive contents of portlandite ($\text{Ca}(\text{OH})_2$) and calcite, observed for >10 wt.% lime, had a slight detrimental effect on the measured physical properties. In all cases, lime amended adobe bricks are suitable for rural building construction in tropical countries.
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1. Introduction

The use of adobe bricks in the construction of shelters in developing countries is mainly dependent on their mechanical resistance, water absorption and chemical behaviours. To improve the physical properties of these building materials, straw debris, fly ash and small amounts of cement or lime are commonly introduced. In spite of the positive impacts of these amendments on the quality of adobe bricks, a little attention has paid to their effects on the microstructure of the traditional building materials.

If the chemical processes and the geotechnical properties of lime-stabilized soils have been investigated [1–8], a little study has been devoted to the assessment of the effects of lime additions on the microstructure and the physical properties of adobe clayey bricks.

The aim of this work is to study the effects of lime additions on the microstructure transformations and the physical properties (bending and compressive strengths, water absorption) of adobe bricks made of a raw clay material from Burkina Faso.

2. Materials

The basic raw material is from Nadialpou (Burkina Faso). Its mineralogical and chemical compositions are given in Table 1.

The chosen lime is a Spanish commercial product, commonly used in the country. As can be derived from Fig. 1, it

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Table 1
Chemical and mineralogical compositions of the concerned raw material

Constituent	Wt. %
SiO ₂	79.36
Al ₂ O ₃	11.85
Fe ₂ O ₃	1.81
MgO	tr
CaO	0.03
CuO	tr
MnO	tr
Na ₂ O	tr
K ₂ O	0.33
TiO ₂	1.27
L.O.I. ^a (1000 °C)	6.27
Kaolinite	30
Quartz	65
Goethite	2
K-feldspar	2

tr – Trace.

^a Loss on ignition.

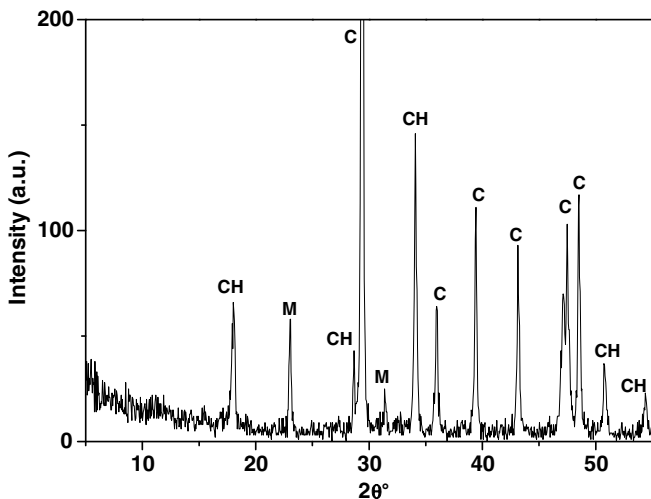


Fig. 1. X-ray diffraction pattern of the used lime. C, CaO; CH, Ca(OH)₂; M, magnesium carbonate hydrate.

consists of CaO(C) and portlandite (Ca(OH)₂ or CH) and a subsidiary compound identified as magnesium carbonate hydrate (M).

3. Experimental methods

Dried samples of the ground raw material, consisting of particles <0.315 mm in size, were mixed with different amounts of lime (up to 12 wt.%) and an optimum amount of water (30 wt.%). The obtained pastes were homogenized for 15 min, manually pressed within prismatic moulds (4 cm × 4 cm × 16 cm) and kept in situ for 24 h. The as-moulded bricks were stored for 30 days at room temperature. This corresponds to the traditionally adopted duration.

The microstructural changes of the adobe bricks were investigated by X-ray diffraction (XRD), infrared spectroscopy (IR), differential thermal analysis (DTA), scanning electron microscopy (SEM) and energy dispersive spec-

troscopy (EDS). The XRD analyses were performed by means of a Philips X'Pert MPD diffractometer equipped with a Cu K α radiation. The IR spectra were obtained by a Nicolet 510FT-IR spectrometer functioning in the range 4000 and 400 cm⁻¹. The thermal curves were obtained by a Labsys-Setaram apparatus, operating at 10 °C/min, in inert atmosphere of (N₂).

Freshly fractured pieces of bricks were carbon coated and examined with a Jeol JSM 5500 scanning electron microscope, equipped with a Falcon EDAX system. The elemental quantitative analyses were performed by EDAX technique, without standards (ZAF method). The maximum resolution turns around 3.5 nm.

The bending strength of adobe bricks was determined according to the NF P14-306 standard [9], using a R.M.U – Testing equipment Bergamo. The operating mode of the latter apparatus is based on the known 3-points technique and the bending strength (R , in MPa) is evaluated by the relation $R = \frac{3}{2} \frac{FL}{b^3}$, where F (in N) is the maximum charge, $L = 106.7$ mm and $b = 40$ mm. The applied charge was 50 N/s. For each run, three specimens were used and the deviation is in the range 0.06–0.14 MPa.

The compressive strength (R_c in MPa) of adobe bricks was calculated using the relation $R_c = \frac{10F}{S}$, where F (in kN) is the maximum charge and S (in cm²) the cross section of the test-piece. For the maximum charge determination, a hydraulic press multitester type Seidner D7940 Riedlingen, operating at 0.025 mm/s is used. The measurement was realized following the NF P14-306 norm [9]. For each run, six half specimens obtained after the bending strength test were considered and the standard variation is in the interval 0.2–0.45 MPa.

The water absorption of adobe bricks was evaluated according to the NF P13-305 standard [10]. Its deviation is <3.5%.

4. Results

4.1. X-ray diffraction

A global analysis of the X-ray diffraction patterns of powdered samples of lime-clayey crude bricks (Fig. 2) showed the presence of kaolinite, quartz, goethite and K-feldspar as well as calcite and portlandite (CH). A comparative examination of the intensity of the main X-ray reflections relevant to the two latter compounds showed that their amounts increased with lime additions, while that of quartz experienced a slight decrease. A careful examination of the X-ray diffraction pattern relevant to the 6 wt.% lime sample exhibited the presence of diffuse and weak reflections at $48^\circ < 2\theta < 50^\circ$. These could be assignable to poorly crystallized silicate calcium hydrate (CSH) [11].

4.2. Infrared spectroscopy

To evidence the presence of non-crystallized phases, crushed samples of bricks were analysed by IR spectrome-

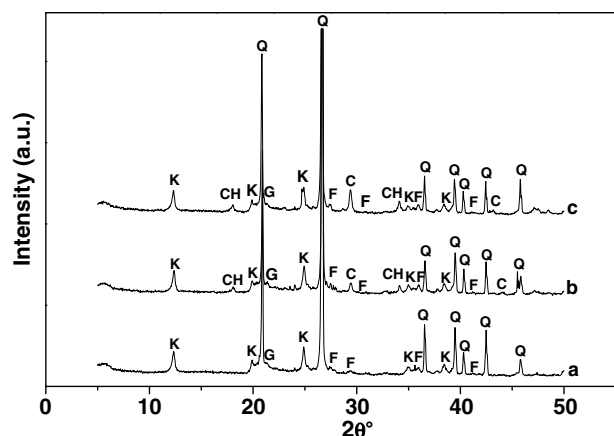


Fig. 2. X-ray diffraction patterns of adobe bricks samples. (a) Raw material, (b) 6 wt.% lime and (c) 10 wt.% lime. K, kaolinite; Q, quartz; G, goethite; F, K-feldspar; CH, portlandite; C, calcite.

try (Fig. 3). The obtained spectra exhibited the prominent bands of quartz (798, 781 and 397 cm^{-1}) [12], hygroscopic water (1640 cm^{-1}) [12] and kaolinite (3700, 3668, 3620, 1090, 1050, 920, 696, 520, 470 cm^{-1}) [12–15]. The latter mineral exhibited a disordered structure since its character-

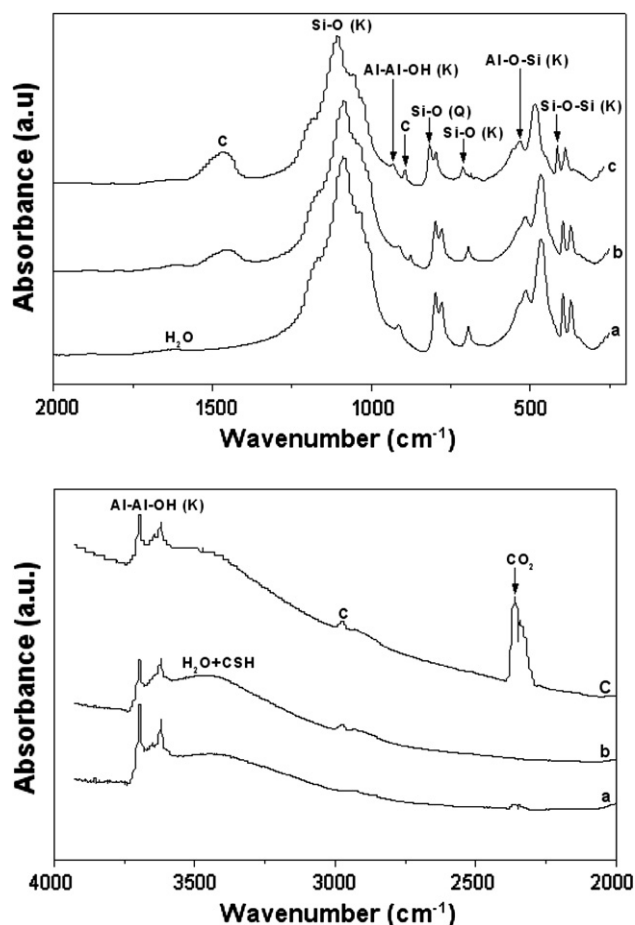


Fig. 3. Infrared spectra of adobe bricks samples. (a) Raw material, (b) 6 wt.% lime and (c) 10 wt.% lime. K, kaolinite; Q, quartz; C, calcite; CSH, calcium silicate hydrate.

istic band at 3672 cm^{-1} is absent [12,15]. The presence of CH and calcite is revealed by the intensification of the band at 3650 cm^{-1} and the occurrence of the bands located at about 879, 1470, 2940, 2980 cm^{-1} [16], respectively. As far as the infrared bands attribution is concerned, one may notice that the intensity of the broad stretching vibration at around 3400 cm^{-1} (Fig. 3b), which is commonly attributed to water hydration, increased while the overtone band at 1650 cm^{-1} decreased. This result lets think that the former band is due to CSH [16] and physisorbed water. The infrared band located at 2350 cm^{-1} (Fig. 3c) is originated from carbon dioxide, which is extensively dissolved in lime-rich samples.

4.3. Differential thermal analyses

Differential thermal analyses performed on the basic material and its lime mixtures (Fig. 4) showed the occurrence of endothermic effects at around 85, 480, 537 and 567 °C corresponding to the loss of hygroscopic water, the dehydroxylation of CH and kaolinite, and to the transformation of α to β quartz, respectively. The exothermic peak manifesting at about 950 °C is due to the structural reorganisation of metakaolinite [17]. The presence of goethite is discriminated by the feeble endotherm at around 348 °C. The broad endothermic phenomenon between 335 and 440 °C (curve b) could be both attributable to the loss of structural water of CSH and goethite [18,19], while the endotherm at about 797 °C is linked to the decomposition of calcite.

4.4. SEM examination and EDS analyses

SEM examinations performed on free lime and 4 wt.% lime samples (Fig. 5a and b) showed a heterogeneous structure consisting of isolated particles of kaolinite

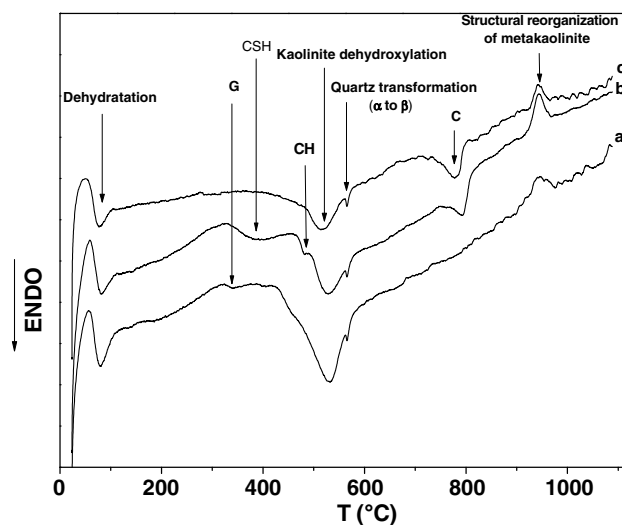


Fig. 4. Differential thermal analyses curves of adobe bricks samples. (a) Raw material, (b) 6 wt.% lime and (c) 10 wt.% lime. G, goethite; C, calcite; CH, portlandite; CSH, calcium silicate hydrate.

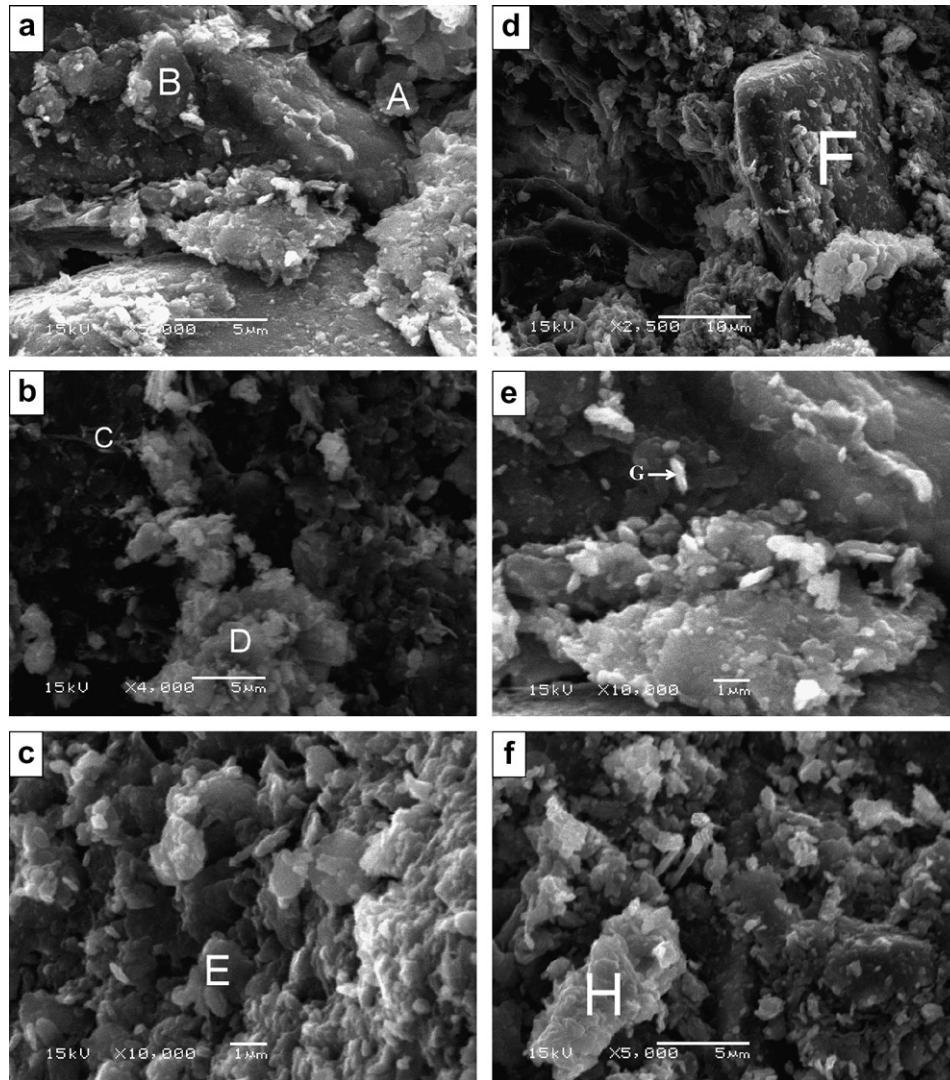


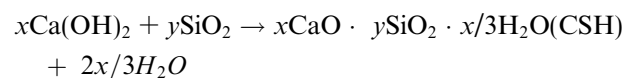
Fig. 5. SEM micrographs of adobe clayey bricks samples. (a) Lime free sample, (b) 4 wt.% lime, (c) 6 wt.% lime, (d and e) 10 wt.% lime and (f) 12 wt.% lime.

(Fig. 6A), coarse euhedral grains of quartz (Fig. 6B) and tiny acicular particles of free silica (Fig. 6C). The structural formula of kaolinite, deduced from its EDS analyses (Fig. 6A) is $\text{Na}_{0.02}(\text{Si}_{1.98}\text{Al}_{0.02})(\text{Al}_{1.69}\text{Fe}_{0.18}\text{Ti}_{0.03}\text{Mg}_{0.1})\text{O}_5(\text{F},\text{OH})_4$. Calcium-rich zones, manifesting as bright areas (Fig. 5b – Zone D), developed in 4 wt.% lime sample. As can be derived from spectrum D of Fig. 6, these areas consisted of a mixture of kaolinite, fine silica and calcic-compounds (calcite and/or CSH). When the lime content increased to 6 wt.%, the tiny filiform free silica disappeared and linkage between particles developed, resulting thus in the occurrence of a homogeneous microstructure (Fig. 5c). In this case, isolated featureless particles (Fig. 5 e-Zone E), identified as CSH (Fig. 6E), were developed. On the other hand, it is worth noting that coarse grains of quartz displayed surficial pits (Fig. 5d-Zone F; Fig. 6F), which apparently constituted preferential sites for CSH and/or calcite nucleation. However, it appears that K-feldspar grains (Fig. 5 e-Zone G; Fig. 6G) were

unaltered. As the lime content increased to 12 wt.%, portlandite and calcite, which manifested as bright areas (Fig. 5 f-Zone H; Fig. 6H), were extensively formed. The excessive carbonation resulted in the occurrence of a heterogeneous microstructure.

5. Discussion

In the light of the above results, it derived that CSH is originated from the reaction between lime and fine grains of quartz. The solubility of fine quartz in the medium is linked to the increase of pH, due to lime additions (Fig. 7). In this condition, the calcium silica hydrate formed according to the reaction:



which is the combination of the following chemical equations:

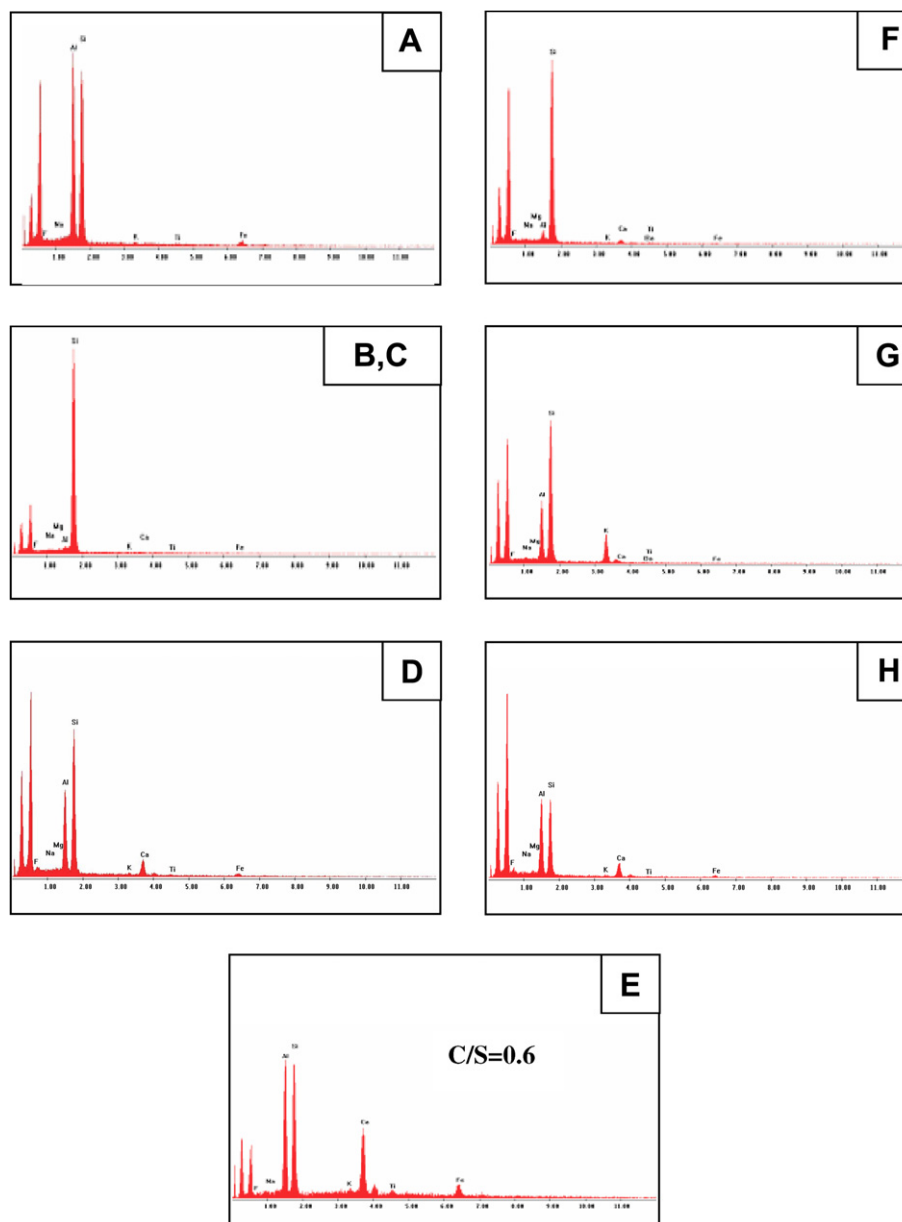
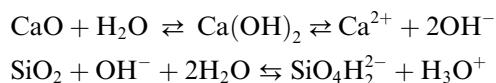


Fig. 6. EDS analyses of areas quoted in Fig. 5.



Referring to the EDS analyses (Fig. 6E), the ratio $x/y = \text{C/S}$ turns around 0.6. This is close to the values reported for CSH [2].

It may be remarked that flocculation–agglomeration of particles, observed for lime–soil mixtures, does not manifest in the present study likely because of the low cation exchange capacity (4.6 meq/100 g) of the concerned material. Moreover, the pozzolanic reaction involving kaolinite seems to be insignificant.

The quantitative formation of calcite in lime-rich samples is the manifestation of the reaction of the hydrated lime with the atmospheric carbon dioxide. The dissolution

of CO_2 within the pastes increased with the increase of calcium affinity and consequently the carbonation reaction became predominant. Apparently, the excessive carbonation of the samples limited the formation of the cementitious hydrated compound.

As a result of lime additions the compressive and bending strengths enhanced and water absorption declined (Figs. 8–10). However, higher lime additions (>10 wt.%) have partial detrimental effects on these properties. In all cases, the magnitudes of strengths are in the range of the required values for adobe bricks [20–22]. The improvement of the physical properties is attributed to the cementitious character of CSH and to the development of minor amounts of calcite and portlandite. The quantitative formation of the latter compounds resulted in the reduction

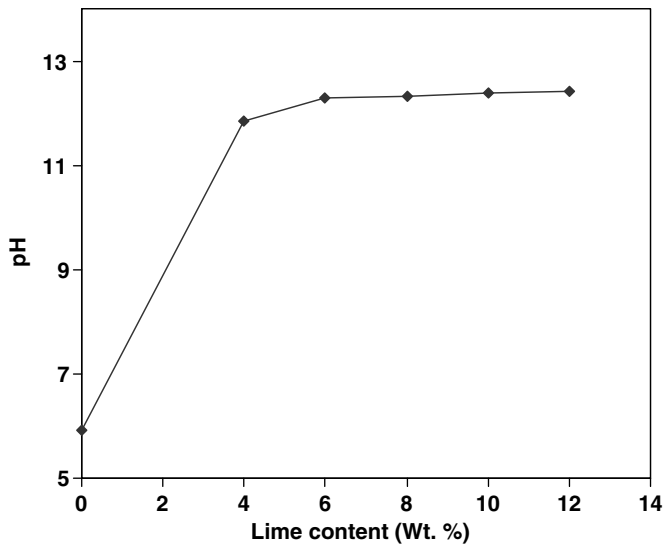


Fig. 7. Evolution of pH with lime additions.

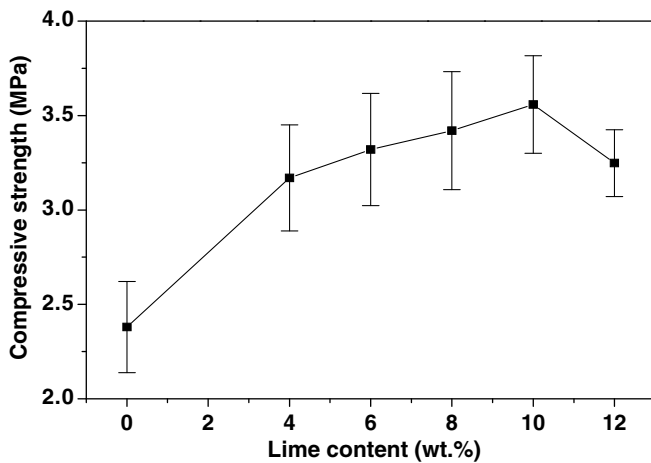


Fig. 8. Evolution of the compressive strength of clayey adobe bricks as function of lime content.

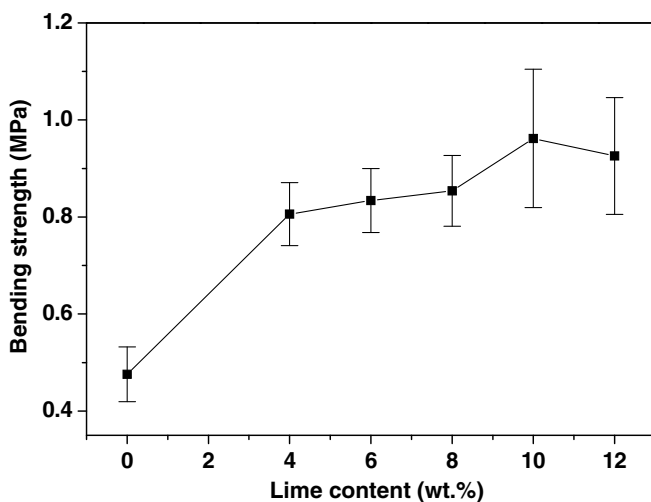


Fig. 9. Variation of the bending strength of adobe bricks versus lime additions.

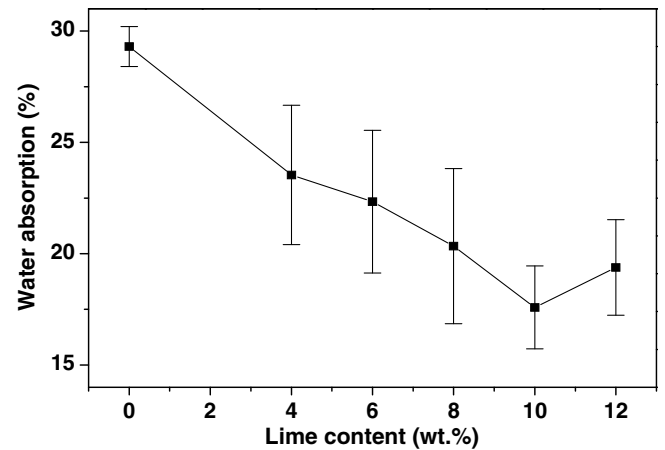


Fig. 10. Effect of lime additions on the evolution of water absorption of adobe bricks.

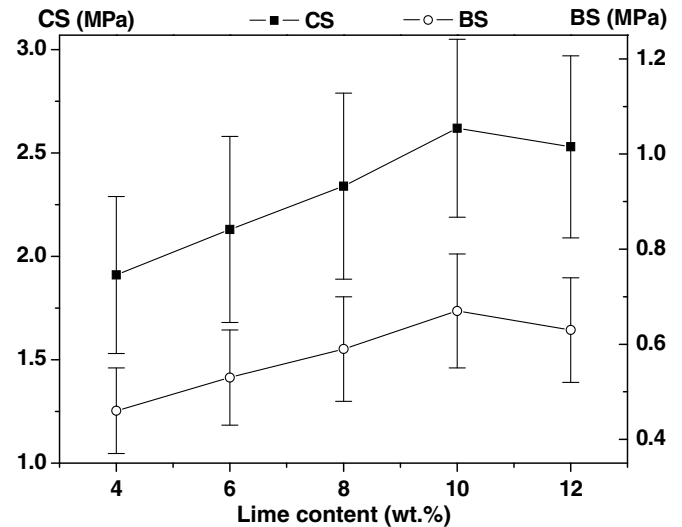


Fig. 11. Change of the compressive and bending strengths (CS and BS) of water immersed adobe bricks against lime admixtures.

of the mechanical resistance and the increase of porosity, characterized by an increase of water absorption.

To test the impact of rain (the main environmental eroding factor in tropical climate) on the adobe bricks, these were totally immersed in water for four days and subjected to compressive and bending strengths (Fig. 11). The results showed that free lime bricks were deteriorated and the obtained values for amended bricks are still close to the standard ones.

6. Conclusion

Lime additions to quartz-rich clayey material for adobe bricks elaboration induced the development of calcite and calcium silicate hydrate, which is mainly formed from the reaction of lime and tiny silica. The presence of the latter compound contributed to the reinforcement of adobe bricks. However, the excessive formation of portlandite and calcite, and the minor development of CSH affected

negatively the mechanical resistance of adobe bricks. Finally, for the elaboration of strengthen and compacted lime-clayey adobe bricks, it may be recommended to reduce the grain size of quartz, increase the duration of hydration and preserve the paste from air.

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